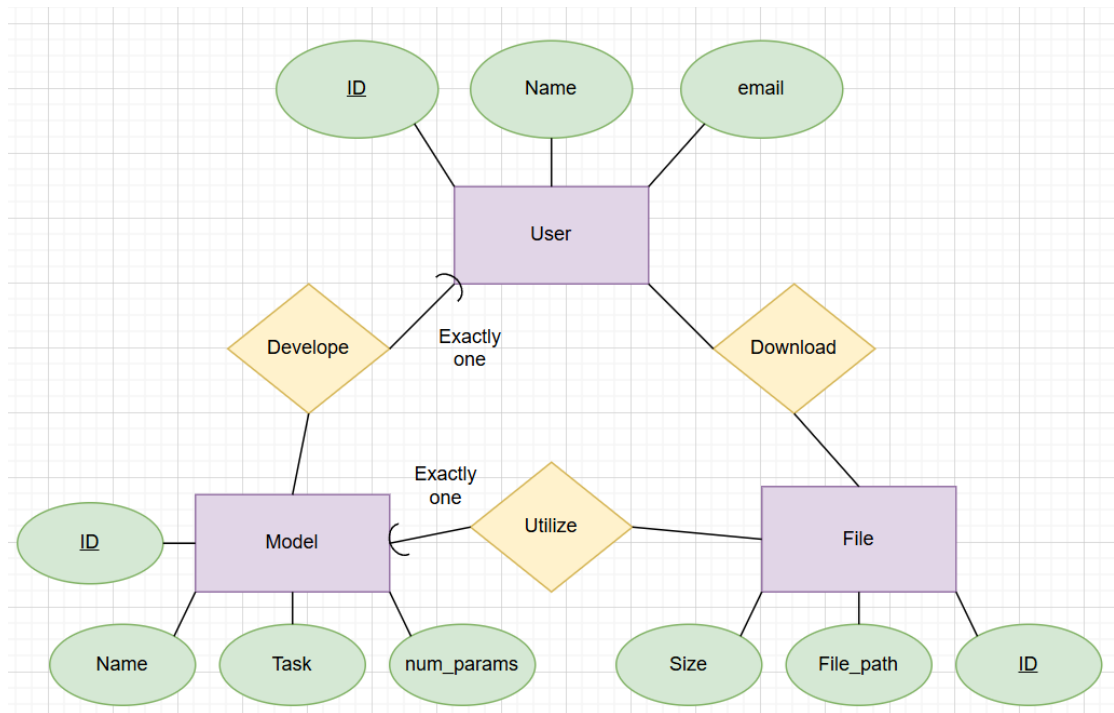


DSCI551 HW3

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1. ER model



2. SQL

(1) Query:

```
select task, count(f.id)
from model as m, file as f
where m.id = f.model_id
group by task
order by count(f.id) desc, m.task asc;
```

Result:

	task	count(f.id)
	language generation	8
	image classification	4
	language understanding	4
	speech recognition	4
	image-text embedding	2
	object detection	2
	semantic segmentation	2
▶	text-to-image	2
	text-to-text generation	2

(2) Query:

```
select task
from model
group by task
having count(id) >= 2
order by task asc;
```

Result:

	task
▶	image classification
	language generation
	language understanding
	speech recognition

(3) Query:

```
select distinct m1.task
from model as m1
join model as m2
on m1.task = m2.task and m1.user_id != m2.user_id
order by m1.task asc;
```

Result:

	task
▶	image classification
	language generation
	language understanding
	speech recognition

(4) Query:

```
select avg(size)
from file as f, model as m
where f.model_id = m.id and m.task = 'speech recognition'
and f.file_path like '%safetensors';
```

Result:

	avg(size)
▶	450

(5) Query:

```
with fd as(
    select file_id, count(user_id) as d_times
    from download
    group by file_id
),
md as(
    select max(d_times) as max_times
    from fd
)
select file_id
from fd, md
```

where fd.d_times = md.max_times

order by file_id asc;

Result:

	file_id
▶	f1
	f11

(6) Query:

select d.user_id, d.file_id

from download as d, file as f, model as m

where d.file_id = f.id and f.model_id = m.id and m.user_id = d.user_id

order by d.user_id asc, d.file_id asc

limit 5;

Result:

	user_id	file_id
▶	u1	f1
	u1	f2
	u10	f19
	u10	f20
	u11	f21

(7) Query:

(select distinct user_id

from download

where file_id = 'f1'

order by user_id asc)

intersect

(select distinct user_id

from download

```
where file_id = 'f2'
```

```
order by user_id asc);
```

Result:

	user_id
▶	u1

Alternative query (if some SQL versions don't support "intersect"):

```
select distinct d1.user_id
```

```
from download as d1
```

```
join download as d2
```

```
on d1.user_id = d2.user_id
```

```
and d1.file_id = 'f1' and d2.file_id = 'f2'
```

```
order by d1.user_id asc;
```

(8) Query:

```
with mp as(
```

```
    select max(num_params) as max_p
```

```
    from model
```

```
    where task = 'language generation'
```

```
)
```

```
select model.name
```

```
from model, mp
```

```
where model.task = 'language generation'
```

```
and mp.max_p = model.num_params;
```

Result:

	name
▶	llama-3